SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – CONCEPT MAP

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 1 – Concept Map
Weightage:	60 % of CIA	**Total Marks:**	20 marks
CO1 Statement:	*Analyse the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
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Section / Batch:	IT-2024	**Date:**	

Code of Conduct

I _Sri Janani B(192421036) (Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Sri Janani B

Instructions

- This test contains ONE question.
- Duration: 30 minutes (At home)

Annexure A - QUESTIONS

1. Construct a concept map that evaluates the interdependence among cellular network architecture, protocol stack, and mobility management.
   Assess how these components collectively improve communication reliability.
   Identify the strongest conceptual relationships.
   Justify your evaluation using appropriate linkages.

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Cellular Network Architecture

Base Station (eNB / gNB)	Core Network (5G CN / EPC)
User Equipment	cellular cell

Provides:
Radio access, coverage & capacity, Handover support, Core network connectivity

Architecture defines available radio resources & network entities for protocol

Protocol stack uses architecture services (eg., radio Interfaces)

protocol Stack
Application layer (apps)
Transport layer (TCP/UDP)
Network layer (IP, routing)
Datalink layer (MAC, RLC)
Physical layer (radio transmission)

Functions:
Data Formatting & addressing,
Error control & Flow control
Secure communication

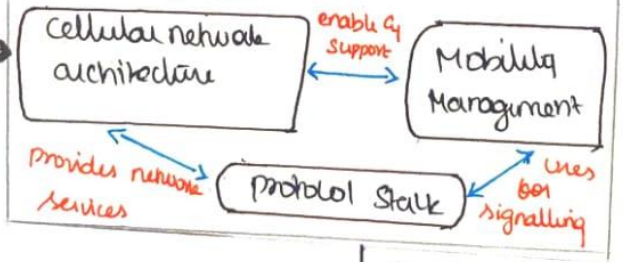
Mobility management uses Protocol Stack

Protocol stack provides signalling to mobility

Mobility Management
Entities Involved
Mobile Terminal
Serving & Target Base station
Mobility Management Entity

Key Function:
Handover, Tracking & location update, session continuity,
Network selection (4G/5G),
Power & resource management.

Independence



Together they ensure.

Improved communication Reliability
Seamless handovers → fewer disconnects
Efficient resource use → better network performance.
Continuous connectivity → reliable voice, data & internet service

Justification.

Architecture gives the physical & logical infrastructure without which mobility & protocol cannot function

Protocol Stack uses services from architecture and is triggered by mobility events.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____